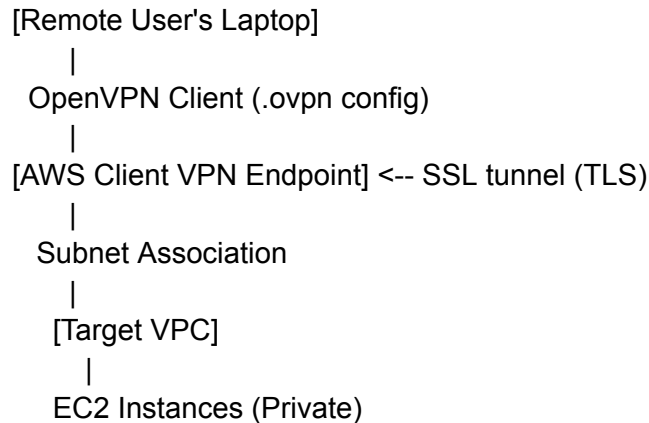

Lab: AWS Client VPN – Step-by-Step (Detailed)

Architecture Overview



◆ Step 1: Create a VPC and Subnet

We'll first create the network where our target resources (like EC2) live.

VPC Details:

- **VPC CIDR Block:** `10.10.0.0/16`
- **Subnet CIDR Block:** `10.10.1.0/24` (in a single AZ, e.g., `us-east-1a`)
- Enable **auto-assign public IP** for easier EC2 testing

You can create these from the AWS Console or use this AWS CLI:

```
aws ec2 create-vpc --cidr-block 10.10.0.0/16
aws ec2 create-subnet --vpc-id <VPC-ID> --cidr-block 10.10.1.0/24 --availability-zone
us-east-1a
```

◆ Step 2: Generate Certificates for Mutual Authentication

AWS Client VPN requires:

- A **server certificate** (to identify AWS endpoint)
- A **client certificate** (to identify client machines)

Use AWS's EasyRSA Tool

1. Clone the repository:

```
git clone https://github.com/OpenVPN/easy-rsa.git
cd easy-rsa/easyrsa3
./easyrsa init-pki
./easyrsa build-ca
```

2. Create Server Certificate:

```
./easyrsa build-server-full server nopass
```

3. Create Client Certificate:

```
./easyrsa build-client-full client1 nopass
```

4. Upload only the **server certificate and chain** to **ACM (N. Virginia)**:

Go to **ACM > Import Certificate** and upload:

- Certificate: `pki/issued/server.crt`
- Private key: `pki/private/server.key`
- CA cert: `pki/ca.crt`

◆ Step 3: Create Client VPN Endpoint

In AWS Console:

Go to **VPC** → **Client VPN Endpoints** → **Create Client VPN Endpoint**

Key Configurations:

- **Client IPv4 CIDR:** 172.16.0.0/22 (must not overlap with VPC)
- **Server certificate ARN:** From ACM
- **Client authentication:** Mutual Authentication
- **Client certificate ARN:** Also from ACM
- **VPC:** Choose the 10.10.0.0/16 VPC
- **Split-Tunnel:** ☒ Yes (means only AWS traffic goes via VPN)

Subnet Association:

- Associate the endpoint with 10.10.1.0/24 subnet
- AWS will automatically create and manage **ENIs (Elastic Network Interfaces)** in that subnet

◆ Step 4: Add Authorization Rule

After creating the VPN endpoint, allow client users to access specific resources:

- Go to the VPN endpoint
- Choose **Authorization Rules**
- Click **Add authorization rule**

- Destination network: `10.10.0.0/16`
- Grant access to: `Allow access to all users`

◆ Step 5: Download and Edit VPN Client Configuration

1. In the VPN Endpoint dashboard, click **Download client configuration** — you'll get a `.ovpn` file.
2. Open it with a text editor, and **insert the client certs** into the file:

```
<cert>
(paste contents of pki/issued/client1.crt)
</cert>
<key>
(paste contents of pki/private/client1.key)
</key>
<ca>
(paste contents of pki/ca.crt)
</ca>
```

Now, the file is ready for OpenVPN.

◆ Step 6: Connect Using OpenVPN Client

1. **Install OpenVPN GUI** (Windows/macOS/Linux)
2. Import the `.ovpn` file
3. Connect to the VPN
4. Once connected, test access to AWS:

 **Testing:**

- Launch an EC2 instance in **10.10.1.0/24** subnet (e.g., Ubuntu with no public IP)
- SSH into the instance from your laptop using the **private IP**:

```
ssh -i my-key.pem ec2-user@10.10.1.X
```

(Assuming you configured the EC2 security group to allow SSH from the VPN CIDR block **172.16.0.0/22**)

Clean-Up Resources

- Delete the VPN endpoint
 - Disassociate subnet
 - Remove ACM certificates
 - Terminate EC2
 - Delete VPC/subnet
-

Summary

Component	Purpose
VPC & Subnet	Your private AWS network
Client VPN Endpoint	Entry point for remote VPN clients
Certificates	Mutual authentication between client/server
Authorization Rule	Defines what IP ranges clients can access
OpenVPN GUI	Tool used to connect from your local device
